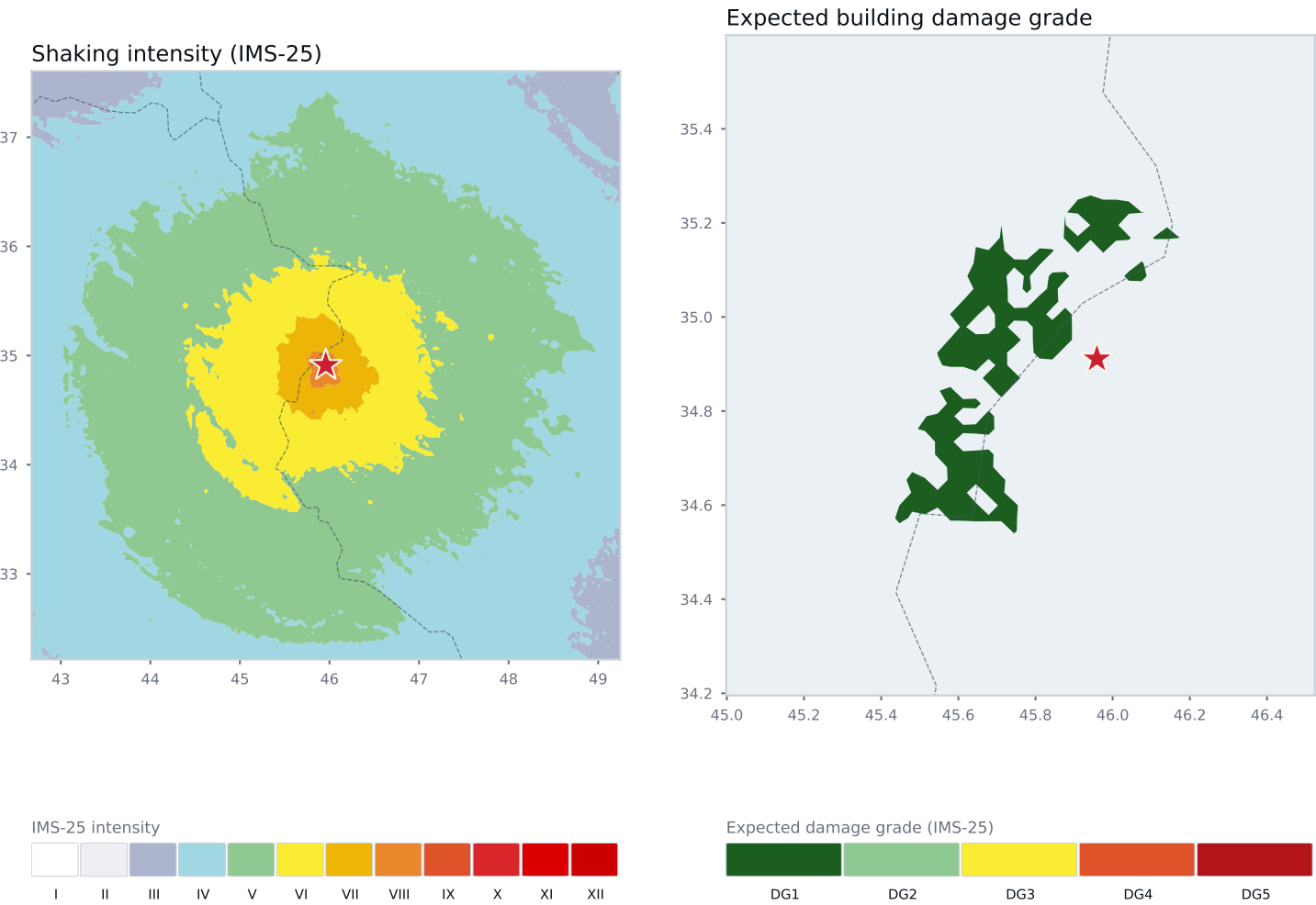




Damage alert

RED

Extensive destruction across the area

Provisional

Transit occupancy

Buildings heavily damaged (DG3 or worse)

Buildings in the shaken area

4.7 thousand to 59 thousand

7.8 million

Central estimate 19 thousand. The range is the 5th to 95th percentile.

21% of them inferred from built-up area, not observed footprints.

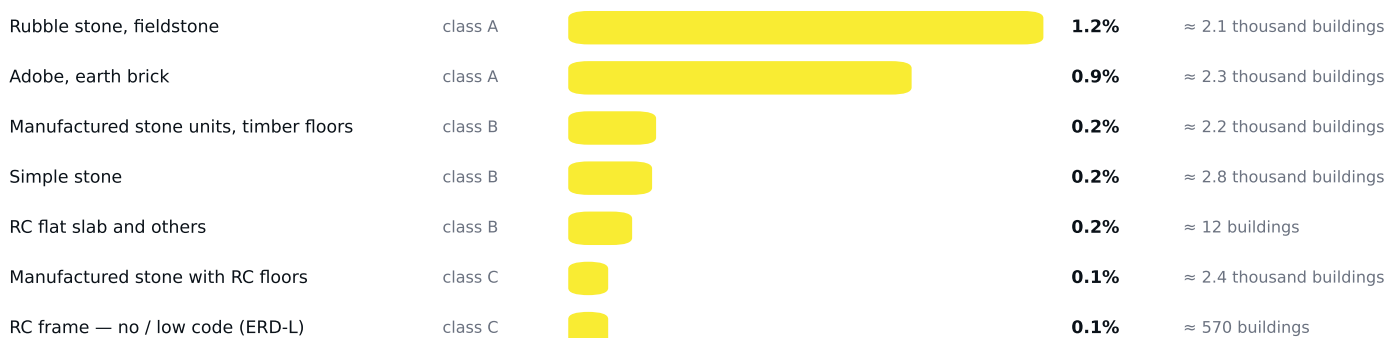


Share of buildings in the mapped area by expected damage

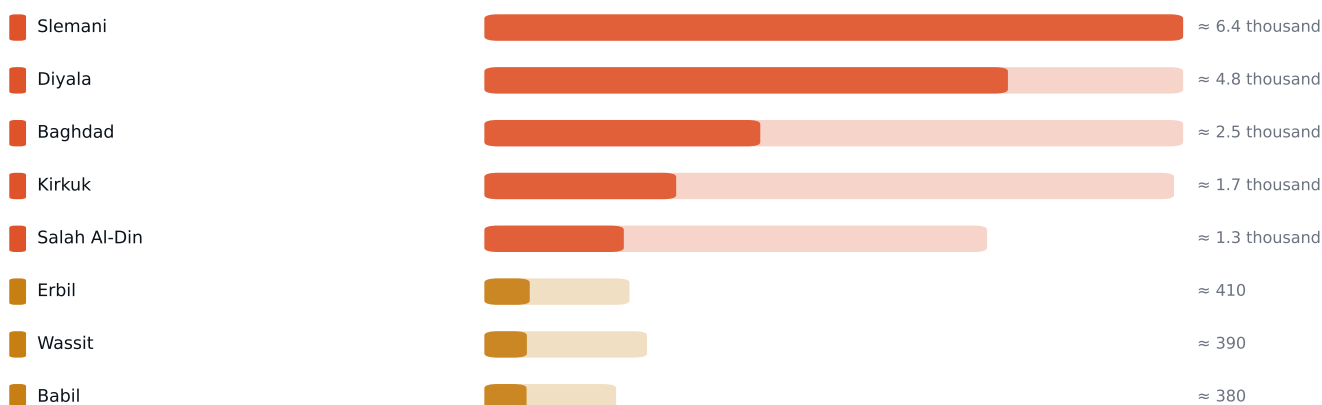


Which buildings failed

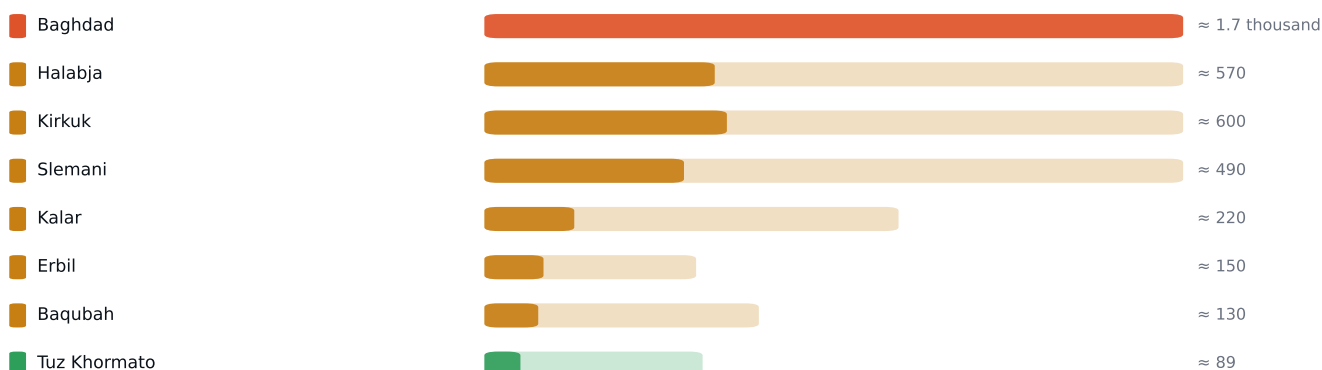
Share of each type's own buildings heavily damaged, worst first



Governorates most affected: heavily damaged buildings, central estimate and 90% range



Cities most affected



Every governorate inside the map

Area	Buildings	Heavy (P50)	90% range	% of its stock
Slemani	668 thousand	4.5 thousand	1.5 thousand to 19 thousand	0.68%
Diyala	605 thousand	3.2 thousand	970 to 15 thousand	0.53%
Baghdad	2.1 million	1.7 thousand	640 to 9.8 thousand	0.08%
Kirkuk	568 thousand	1.1 thousand	310 to 6.3 thousand	0.20%
Salah Al-Din	592 thousand	810	240 to 4.6 thousand	0.14%
Erbil	659 thousand	300	120 to 1.3 thousand	0.05%
Wassit	376 thousand	270	100 to 1.5 thousand	0.07%
Babil	534 thousand	260	91 to 1.2 thousand	0.05%
Ninewa	731 thousand	250	100 to 1.1 thousand	0.03%
Al-Anbar	359 thousand	160	64 to 840	0.04%
Kerbala	314 thousand	100	36 to 420	0.03%
Duhok	234 thousand	28	11 to 120	0.01%
Maysan	17 thousand	8	2 to 41	0.05%
Al-Najaf	19 thousand	5	2 to 24	0.03%

How this report was made

Hazard engine	Bumelerze SHAKEmap 26.1.1, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)
Conditioning	mvn (Engler et al. 2022) on dyfi with reverse-GMICE sigma (UQ-B, D46)
Observations	0 seismic stations, 1131 felt-report cells (felt channel: dyfi)
Distances	finite-fault; rupture: Hayes, personal communication
Damage scale	IMS-25
Provenance	config 09a6a04fa72f, product version 1, event bml20170001

Data sources

Earthquake parameters	USGS, EMSC, GEOFON
Building footprints	GFZ OpenBuildingMap, from OpenStreetMap (c) OpenStreetMap contributors, ODbL; Microsoft Building Footprints, CDLA-Permissive-2.0
Built-up surface, settlements	European Commission JRC, Global Human Settlement Layer, CC BY 4.0
Population	WorldPop, University of Southampton, CC BY 4.0
Administrative boundaries	OCHA COD-AB Iraq, CC BY-IGO; source authority Iraq Central Statistical Organisation
Building types, vulnerability	GEM Global Exposure Model; IMS-25 vulnerability table
Casualty rates (computed, not published)	FEMA Hazus Earthquake Model Technical Manual 5.1

What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Bumelerze SHAKEmap and damage report, event bml20170001, version 1, generated 21 Sep 2026, 22:25 UTC. CC BY 4.0.